

Exam Prep AI - Focused Notes

Generated on 08 June 2026

EXAM PREPARATION NOTES

Concepts:

1. Jigsaw Puzzle

- Definition: A jigsaw puzzle is a problem that requires breaking down a complex task into smaller, manageable pieces.
- Key Points:
 - Divides a problem into smaller, independent sub-problems.

- EACH SUB-PROBLEM IS SOLVED INDEPENDENTLY.

- Solutions to sub-problems are combined to form a complete solution.
- Important Features:
 - Decomposition of a complex problem into smaller sub-problems.

- INDEPENDENT SOLUTION OF EACH SUB-PROBLEM.

- Advantages and Disadvantages:
 - Advantages: Effective for solving complex problems, reduces the complexity of the problem.
 - Disadvantages: May not work well if sub-problems are highly interdependent.
- Common Exam Questions:
 - A jigsaw puzzle is an example of a problem-solving strategy that involves breaking down a complex task into smaller, independent sub-problems. Explain how this strategy works.
 - A company is facing a complex problem that requires breaking down into smaller sub-problems. Which problem-solving strategy would you recommend?

2. Fibonacci Sequence

- Definition: A Fibonacci sequence is a series of numbers in which each number is the sum of the two preceding numbers.
- Key Points:
 - Each number in the sequence is the sum of the two preceding numbers.

- THE SEQUENCE STARTS WITH 0 AND 1.

- Important Features:
 - Each number in the sequence is the sum of the two preceding numbers.
- Advantages and Disadvantages:
 - Advantages: Useful in modeling real-world problems, such as population growth and financial transactions.
 - Disadvantages: May not be useful for all types of problems.
- Common Exam Questions:
 - Explain the Fibonacci sequence and its significance in problem-solving.
 - A company is experiencing rapid growth, and you are asked to model its growth using a sequence. Which sequence would you recommend?

3. Palindrome

- Definition: A palindrome is a word, phrase, or sequence that reads the same backward as forward.
 - Key Points:
 - A palindrome reads the same backward as forward.
 - Palindromes can be used to create symmetry and balance in problem-solving.
 - Important Features:
-

- SYMMETRY AND BALANCE.

- Advantages and Disadvantages:
 - Advantages: Useful in creating symmetry and balance in problem-solving.
 - Disadvantages: May not be useful for all types of problems.
- Common Exam Questions:
 - Explain the concept of a palindrome and its significance in problem-solving.
 - A problem requires symmetry and balance. Which problem-solving strategy would you recommend?

4. Heuristics Approach

- Definition: A heuristics approach is a problem-solving strategy that involves using mental shortcuts to make decisions quickly.
 - Key Points:
 - Involves using mental shortcuts to make decisions.
 - Heuristics can be used when there is limited time or information.
 - Important Features:
-

- MENTAL SHORTCUTS.

- LIMITED TIME OR INFORMATION.

- Advantages and Disadvantages:
 - Advantages: Useful when time is limited or information is scarce.
 - Disadvantages: May not be accurate or reliable.
- Common Exam Questions:
 - Explain the heuristics approach and its significance in problem-solving.
 - A company is facing a tight deadline, and you need to make a quick decision. Which problem-solving strategy would you recommend?

5. Means-Ends Analysis

- Definition: Means-ends analysis is a problem-solving strategy that involves identifying the goal and then determining the means to achieve it.
- Key Points:
 - Identifies the goal and then determines the means to achieve it.
 - Involves breaking down the problem into smaller sub-problems.
- Important Features:
 - Breaking down the problem into smaller sub-problems.
- Advantages and Disadvantages:
 - Advantages: Effective for solving complex problems.
 - Disadvantages: May not work well if sub-problems are highly interdependent.
- Common Exam Questions:
 - Explain the means-ends analysis and its significance in problem-solving.
 - A company is facing a complex problem that requires breaking down into smaller sub-problems. Which problem-solving strategy would you recommend?

6. Backtracking Strategy

- Definition: Backtracking strategy is a problem-solving strategy that involves trying different solutions and then going back to try another solution if the current one does not work.
- Key Points:

- INVOLVES TRYING DIFFERENT SOLUTIONS.

- If the current solution does not work, goes back to try another solution.
- Important Features:

- TRYING DIFFERENT SOLUTIONS.

- GOING BACK TO TRY ANOTHER SOLUTION.

- Advantages and Disadvantages:
 - Advantages: Effective for solving complex problems.
 - Disadvantages: May be time-consuming and may not work well if the problem has multiple solutions.
- Common Exam Questions:
 - Explain the backtracking strategy and its significance in problem-solving.
 - A company is facing a complex problem that requires trying different solutions. Which problem-solving strategy would you recommend?

7. Randomized Approach

- Definition: Randomized approach is a problem-solving strategy that involves using random numbers or variables to solve a problem.
- Key Points:

- USES RANDOM NUMBERS OR VARIABLES.

- Can be used when there is limited information or time.
- Important Features:

- RANDOM NUMBERS OR VARIABLES.

- Advantages and Disadvantages:
 - Advantages: Useful when there is limited information or time.
 - Disadvantages: May not be accurate or reliable.
- Common Exam Questions:
 - Explain the randomized approach and its significance in problem-solving.
 - A company is facing a problem that requires using random numbers or variables. Which problem-solving strategy would you recommend?

8. Grading System

- Definition: Grading system is a problem-solving strategy that involves assigning grades or scores to students based on their performance.
- Key Points:
 - Assigns grades or scores to students based on their performance.

- CAN BE USED TO EVALUATE STUDENT PERFORMANCE.

- Important Features:
-

- ASSIGNING GRADES OR SCORES.

- Advantages and Disadvantages:

- Advantages: Useful for evaluating student performance.

- Disadvantages: May not be fair or accurate.

- Common Exam Questions:

- Explain the grading system and its significance in problem-solving.

- A school is implementing a grading system to evaluate student performance. Which problem-solving strategy would you recommend?

9. Delivery Times

- Definition: Delivery times are the time it takes for a product to be delivered to the customer.

- Key Points:

- TIME IT TAKES FOR A PRODUCT TO BE DELIVERED.

- Can be used to evaluate logistics and supply chain management.

- Important Features:

- TIME IT TAKES FOR A PRODUCT TO BE DELIVERED.

- Advantages and Disadvantages:

- Advantages: Useful for evaluating logistics and supply chain management.

- Disadvantages: May not be accurate or reliable.

- Common Exam Questions:

- Explain the concept of delivery times and their significance in problem-solving.

- A company is facing a problem with delivery times. Which problem-solving strategy would you recommend?

10. Sales Figures

- Definition: Sales figures are the amount of money earned by a company through sales.

- Key Points:

- AMOUNT OF MONEY EARNED THROUGH SALES.

- CAN BE USED TO EVALUATE BUSINESS PERFORMANCE.

- Important Features:

- AMOUNT OF MONEY EARNED THROUGH SALES.

- Advantages and Disadvantages:

- Advantages: Useful for evaluating business performance.

- Disadvantages: May not be accurate or reliable.

- Common Exam Questions:

- Explain the concept of sales figures and their significance in problem-solving.

- A company is facing a problem with sales figures. Which problem-solving strategy would you recommend?

11. Loan Interest Rates

- Definition: Loan interest rates are the interest charged on a loan.

- Key Points:

- INTEREST CHARGED ON A LOAN.

- CAN BE USED TO EVALUATE FINANCIAL PERFORMANCE.

- Important Features:
-

- INTEREST CHARGED ON A LOAN.

- Advantages and Disadvantages:
 - Advantages: Useful for evaluating financial performance.
 - Disadvantages: May not be accurate or reliable.
 - Common Exam Questions:
 - Explain the concept of loan interest rates and their significance in problem-solving.
 - A company is facing a problem with loan interest rates. Which problem-solving strategy would you recommend?
12. Text Processing
- Definition: Text processing is the process of analyzing and manipulating text data.
 - Key Points:
-

- ANALYZING AND MANIPULATING TEXT DATA.

- CAN BE USED TO EVALUATE TEXT-BASED INFORMATION.

- Important Features:
-

- ANALYZING AND MANIPULATING TEXT DATA.

- Advantages and Disadvantages:
 - Advantages: Useful for evaluating text-based information.
 - Disadvantages: May not be accurate or reliable.
 - Common Exam Questions:
 - Explain the concept of text processing and its significance in problem-solving.
 - A company is facing a problem with text-based information. Which problem-solving strategy would you recommend?
13. Content Moderation
- Definition: Content moderation is the process of monitoring and controlling online content.
 - Key Points:
-

- MONITORING AND CONTROLLING ONLINE CONTENT.

- CAN BE USED TO EVALUATE ONLINE BEHAVIOR.

- Important Features:
-

- MONITORING AND CONTROLLING ONLINE CONTENT.

- Advantages and Disadvantages:

- Advantages: Useful for evaluating online behavior.
- Disadvantages: May not be accurate or reliable.
- Common Exam Questions:
- Explain the concept of content moderation and its significance in problem-solving.

- A COMPANY IS FACING
